

Vehicle to Grid (V2G) and Grid to Vehicle (G2V)

Operation in Electric Vehicle Based Power Systems

Abstract - The increasing number of electric vehicles has created additional stress on existing power grids due to uncontrolled charging during peak hours. In conventional grid to vehicle systems, which leads to inefficient use of available energy resources. This work focus on a vehicle to grid and grid to vehicle operation that enables two-way power flow between electric vehicles and the utility grid. In the proposed approach, Electric vehicles batteries are used as temporary energy storage units that can either draw power from the grid or supply power back when required. A basic control strategy is considered to manage charging and discharging based on load demand conditions. The system performance is analyzed through simulation, and the results show a noticeable reduction in peak load and improved balance in power demand. This study highlights the potential of V2G-G2V technology to support grid stability and efficient energy utilization in future electric power systems.